

RFR Regime

(8. Multi Currency Curves)

FX Quote



- ▶ Understanding FX Quote is crucial to get the conversion rate right. The first currency is called the **base** currency, and the second currency is called the **quote** currency.

$$\begin{array}{ccc} \text{EUR} & / & \text{USD} = 1.1250 \\ \underbrace{\quad} & & \underbrace{\quad} \\ \text{(base currency)} & & \text{(quote currency)} \end{array} \quad (1 \text{ EUR} = 1.1250 \text{ USD})$$

- ▶ There are two popular Quote Conventions. The FX spot and futures can be quoted in different conventions for the same currency pair.

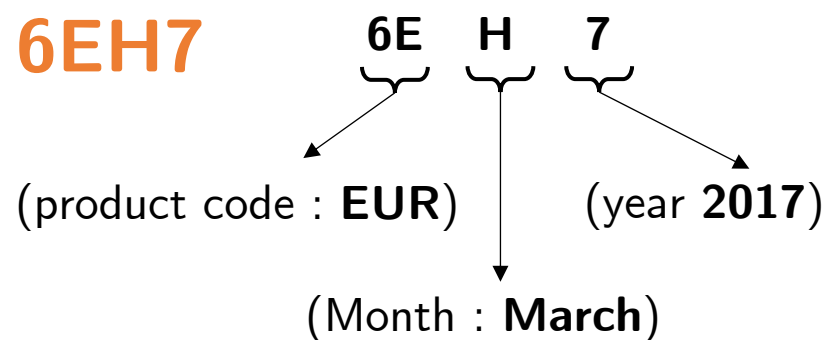
▶ European Quote **USD / EUR , USD / JPY , USD / CHF** (USD = base currency)

▶ American Quote **EUR / USD , JPY / USD , CHF / USD** (USD = quote currency)

CME FX Futures Product Code and Quotes



- ▶ All currency futures on CME Globex are listed using the following trading code convention.



JAN	FEB	MAR	APR	MAY	JUN
F	G	H	J	K	M
JUL	AUG	SEP	OCT	NOV	DEC
N	Q	U	V	X	Z

- ▶ For example, the EUR/USD is quoted as USD per EURO on CME's website

<https://www.cmegroup.com/markets/fx/g10/euro-fx.quotes.html>

<https://www.cmegroup.com/rulebook/CME/>

CME FX Futures Contract Specification

	EUR/USD	JPY/USD	GBP/USD	CAD/USD	AUD/USD	MXN/USD
PRODUCT	Euro	Japanese Yen	British Pound	Canadian Dollar	Australian Dollar	Mexican Peso
PRODUCT SYMBOL	6E	6J	6B	6C	6A	6M
CONTRACT SIZE	125,000 euro	12.5 million yen	62,500 pounds	100,000 Canadian dollars	100,000 Australian dollars	500,000 pesos
EXPIRATIONS	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly, monthly	Quarterly
MINIMUM TICK SIZE	\$0.00005 per euro increments	\$0.0000005 per yen increments	\$0.0001 per pound increments	\$0.00005 per CAD increments	\$0.00005 per AUD increments	\$0.00001 per peso increments
DOLLAR VALUE OF A TICK	\$6.25	\$6.25	\$6.25	\$5.00	\$5.00	\$5.00

Dollar value of a tick = Contract size x Minimum Tick size

Notional Value = Contract size x Quote

PnL = Contract size x (Δ Quote)

FX MARKET SNAPSHOT >

6EM5 OPT  Euro FX Futures 1.1397 169,781 -0.00175 (-0.15%)	6JM5 OPT  Japanese Yen Futures 0.006998 173,554 -0.0000485 (-0.69%)
6BM5 OPT  British Pound Futures 1.3313 63,505 -0.0016 (-0.12%)	6CM5 OPT  Canadian Dollar Futures 0.7237 48,315 +0.0003 (+0.04%)
6MM5 OPT  Mexican Peso Futures 0.050930 29,524 +0.000230 (+0.45%)	6NM5 OPT  New Zealand Dollar Futures 0.5971 24,732 -0.00295 (-0.49%)

CME FX Futures Settlement



- ▶ Standard Futures settle on the **third Wednesday** (IMM dates) and trading terminates the **second business day preceding the third Wednesday** of the contract month (usually Monday)
- ▶ The highest volumes and tightest liquidity typically occur in the nearest quarterly contract month
- ▶ As the last trading day approaches, a trader typically has three choices
 - ▶ Close out the position by entering an opposite trade leaving with no Euro exposure
 - ▶ Roll the future to next expiring quarterly contract month to maintain the EUR exposure. For example, if a trader is long March contract, they can sell **Mar/Jun spread** (sell Mar, buy Jun) to create a long Jun position.
 - ▶ Make Delivery (Short) or Take delivery (long) of the currencies. For example, in case of EUR futures, short positions will deliver 125,000 euros to the long position per standard contract

GBP Futures Settlement Example



- ▶ March contract **6BH5** has Expiry Date = 19th March, Last Trading Date = 17th March
- ▶ Monday (19th March) **6BH5** quote = 1.2405 and deferred contract June **6BM5** quote = 1.2395
- ▶ **Calendar spread = Contract Quote – Deferred Quote = 1.2405 - 1.2395 = 0.0010**
- ▶ In the final 30 seconds of trading CME calculates VWAP of the deferred contract (assume 1.2390)
- ▶ Final Settlement price = 1.2390 + 0.0010 = 1.2400

FX Link



- ▶ CME FX link provides a seamless connection between the OTC spot market and the CME future market. The FX link is a spread between FX Future and FX Spot. This is equivalent to the **Forward points or swap points** which can be +ve or -ve.

$$\text{FX Link Spread} = \text{CME Future price} - \text{OTC Spot price}$$

- ▶ If the future and spot quotes follow the same convention, the FX link is called **non-inverted**, else it's called **inverted**. It's an important consideration to get the calculations right.
- ▶ For **non-inverted** currencies, **long** on the FX link spread means **buying futures** and **selling spot**. For **inverted** currencies, **long** on FX link spread means **buying spot** and **selling futures**.

Covered Interest Parity (CIP)

- ▶ CIP is like a physical law in international finance which works on no arbitrage principle. The **implied future** price is linked to the spot price through the short-term interest rate differential of the two currencies and the days until expiry. This result is called **Covered Interest rate parity** (CIP).

$$\text{Future Price}(F) = \text{Spot price}(S) \times \left(\frac{1 + R_{\text{quote}} \left(\frac{d}{360} \right)}{1 + R_{\text{base}} \left(\frac{d}{360} \right)} \right)$$

- ▶ If $R_{\text{quote}} < R_{\text{base}}$ (positive carry), Then $F < S$ (Future trades at discount) and basis < 0
- ▶ if $R_{\text{quote}} > R_{\text{base}}$ (negative carry) , Then $F > S$ (Future trades at premium) and basis > 0

xCy basis



- ▶ Due to market frictions and specific demand and supply conditions, CIP does not hold in practice. The traded future prices will be slightly different from what is expected from CIP. The difference between the traded price and the CIP price is called as **cross-currency basis**.
- ▶ The CME EUR/USD Cross Currency Basis Index (**XEURB**) measures the basis point deviation from the covered interest parity implied forward FX price. This is the difference between the next ESTR quarterly IMM futures interest rate and the implied ESTR interest rate from the next Euro FX Future and SOFR quarterly IMM futures. It can be calculated using the formula below

$$\text{xCy basis} = \underbrace{\left(\frac{360}{d}\right) \left[\left(\frac{S}{F}\right) \left(1 + R_{SOFR} \frac{d}{360}\right) - 1 \right]}_{\text{(implied ESTR from FX Future)}} - \underbrace{R_{ESTR}}_{\text{(observed ESTR)}}$$

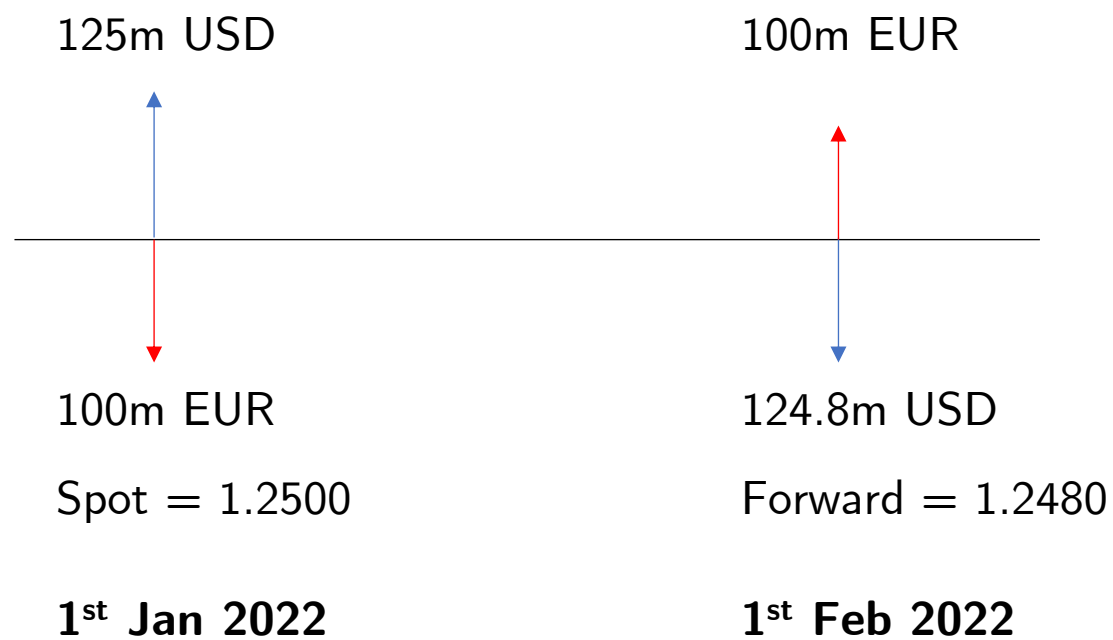
<https://www.cmegroup.com/markets/interest-rates/stirs/eur-xccy.settlements.html>

<https://www.cmegroup.com/markets/interest-rates/cme-group-cross-currency-basis-watch.html>

FX Swap

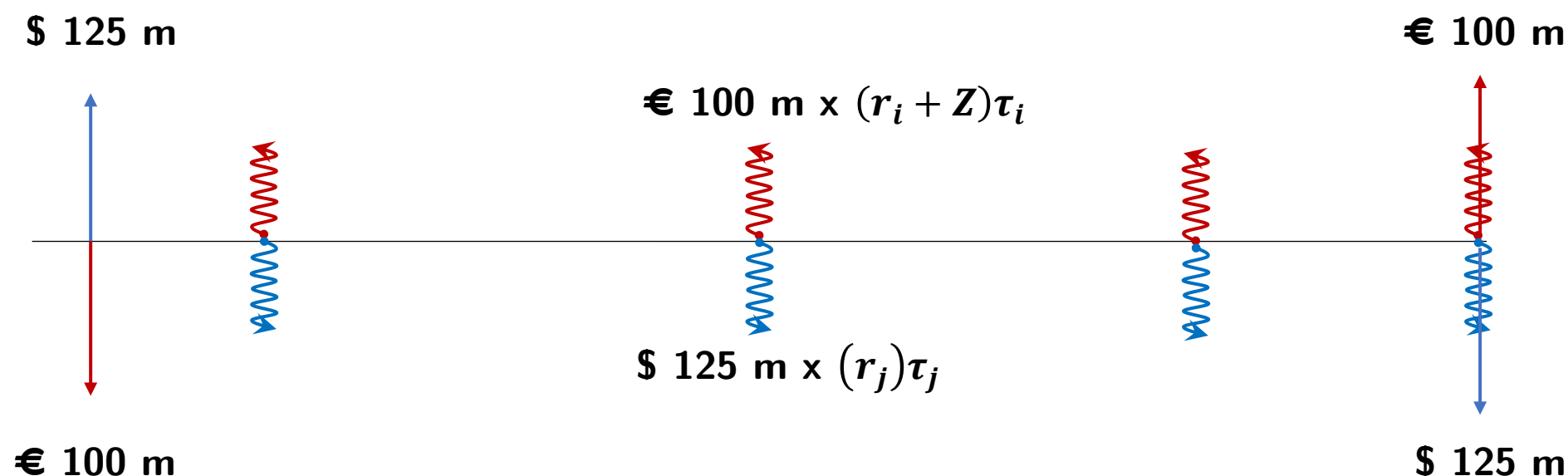


- ▶ FX Swaps are agreements to complete two offsetting exchanges: the "near leg" is the initial exchange of two currencies at the spot rate, and the "far leg" is the reverse exchange of the same currencies at a predetermined forward rate.



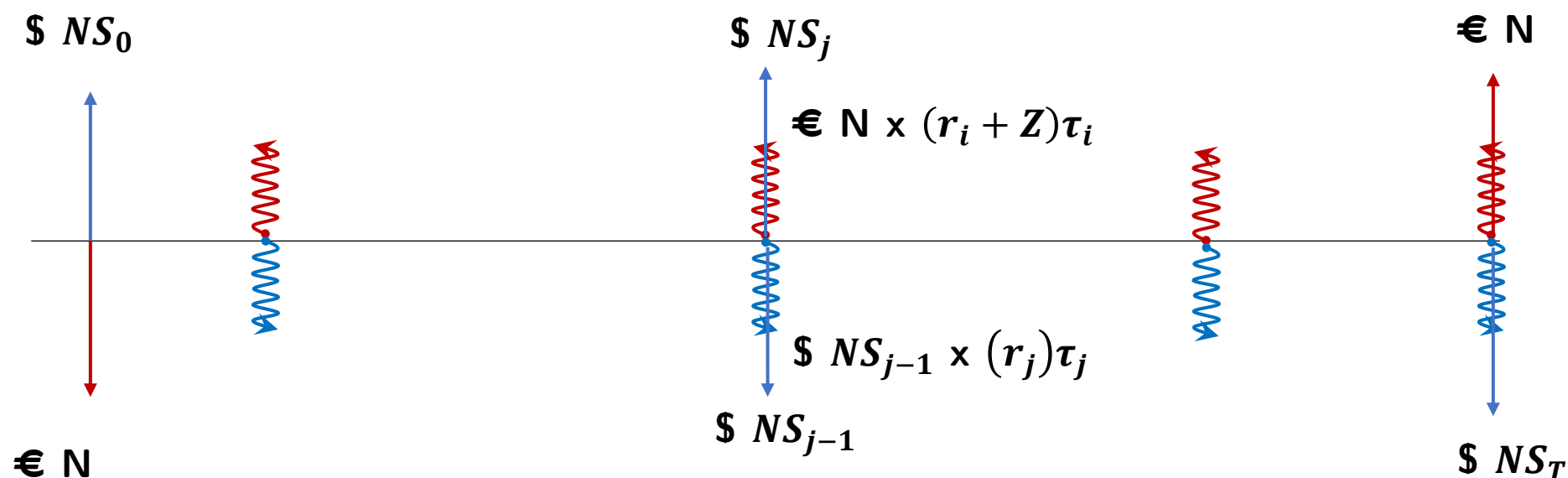
Non-MTM XCCY

- A non-MTM cross-currency swap involves exchanging nationals in two different currencies at the beginning and at the end using the same rate as the beginning and periodic interest payments (fixed or floating). The contract can be customized to have **fixed-fixed**, **fixed-floating** or **floating-floating** payments. Below is a schematic for a non-MTM swap. We typically add a spread(Z) to the non-USD leg for the swap to trade at par.



MTM XCCY

- ▶ An MTM cross-currency swap is the most popular variant of cross-currency swaps. Typically, the non-USD notional is kept fixed, whereas the USD notional gets adjusted based on prevailing spot rates on coupon dates. This is to limit the exposure and credit risk.



Bloomberg MTM XCCY Swap

91) Actions ▾ 92) Products ▾ 93) Views ▾ 94) Info ▾ 95) Settings ▾

Solver (Leg 2: Spread) ▾ Load Save Trade ▾

3) Main 4) Details 5) Curves 6) Cashflow 7) Resets 9) Scenario 10) Risk 12) Matrix

Deal MTM XCCY Swap Counterparty SWAP CNTRPARTY + Ticker / SWAP
CCP OTC Trade Date

Swap

*Notional Reset b...		Arrears	
Leg 1: Float	Receive	Leg 2: Float	Pay
Notional	10,000,000	Notional	9,455,102.45
Currency	USD	Currency	EUR
Effective	0D 10/20/2023	Effective	0D 10/20/2023
Maturity	5Y 10/20/2028	Maturity	5Y 10/20/2028
Index	1D SOFRRATE	Index	1D ESTRON
Spread	0.000 bp	Spread	-24.500 bp
Leverage	1.00000	Leverage	1.00000
Reset Freq	Daily	Reset Freq	Daily
Pay Freq	Quarterly	Pay Freq	Quarterly
Day Count	ACT/360	Day Count	ACT/360

Valuation Setting

Curve Date
Valuation
CSA Coll Ccy
Coll Crv 490
Valuation Ccy
FX Rate
☒ OIS DC Stripping

Market

Dscnt 490 ▾ M ▾ USD SOFR (ICVS) Dscnt 403 ▾ MBB USD Coll for EUR
Fwd 490 ▾ M ▾ USD SOFR (ICVS) Fwd 514 ▾ M ▾ EUR OIS ESTR (I ▾)

Leg 1: NPV 9,999,385.26 Leg 2: NPV -9,999,385.26

Accrued	Premium	DV01
0.00	99.99	2.46
0.00	-99.99	-2.46

Valuation Results

Principal	Accrued	NPV	Premium	BP Value	Gamma (1bp)
0.00	0.00	0.00	0.00000	0.00000	

22) Calculators ▾

EUXOQQ5 ↓ -24.5000 +.5000 BGN -26.3082 / -22.6918 BGN
At 11:11 Op -25.0000 Hi -24.5000 Lo -25.0007 Close -25.0000

EUXOQQ5 BGN Currency Security Description: Swap

Properties Related Instruments Related Curves

EUR USD Basis Swap ESTRON vs SOFRRATE 5Year FIGI BBG00Y8WSTY5

A swap with two floating legs that involves making a payment of one index versus receiving payment in another index. Basis swaps can involve two different indexes in the same currency or indexes in different currencies.

Overview		Float Leg		Float Leg	
Currency	EUR	Day Count	ACT/360	Day Count	ACT/360
Effective Date	20-OCT-2023	Pay Freq	Quarterly	Pay Freq	Quarterly
Maturity Date	20-OCT-2028	Index	ESTRON Index	Index	SOFRRATE Index
Settlement	T+2 Days	Reset Freq	Daily	Reset Freq	Daily
Term	5 Year(s)	Bus Adj	ModifiedFollowing	Bus Adj	ModifiedFollowing
Quote	-24.5 bp	Adjust	Accr and Pay Dates	Adjust	Accr and Pay Dates
FX Index	EURUSD	Roll Conv	Backward (EOM)	Roll Conv	Backward (EOM)
FX Fixing Lag	2 Business Day(s)	Calc Cal	TE, FD	Calc Cal	TE, FD
FX Fixing Cal	EN	Fixing Cal	TE	Fixing Cal	GT

Price Chart | GP »

Last Price -24.50

